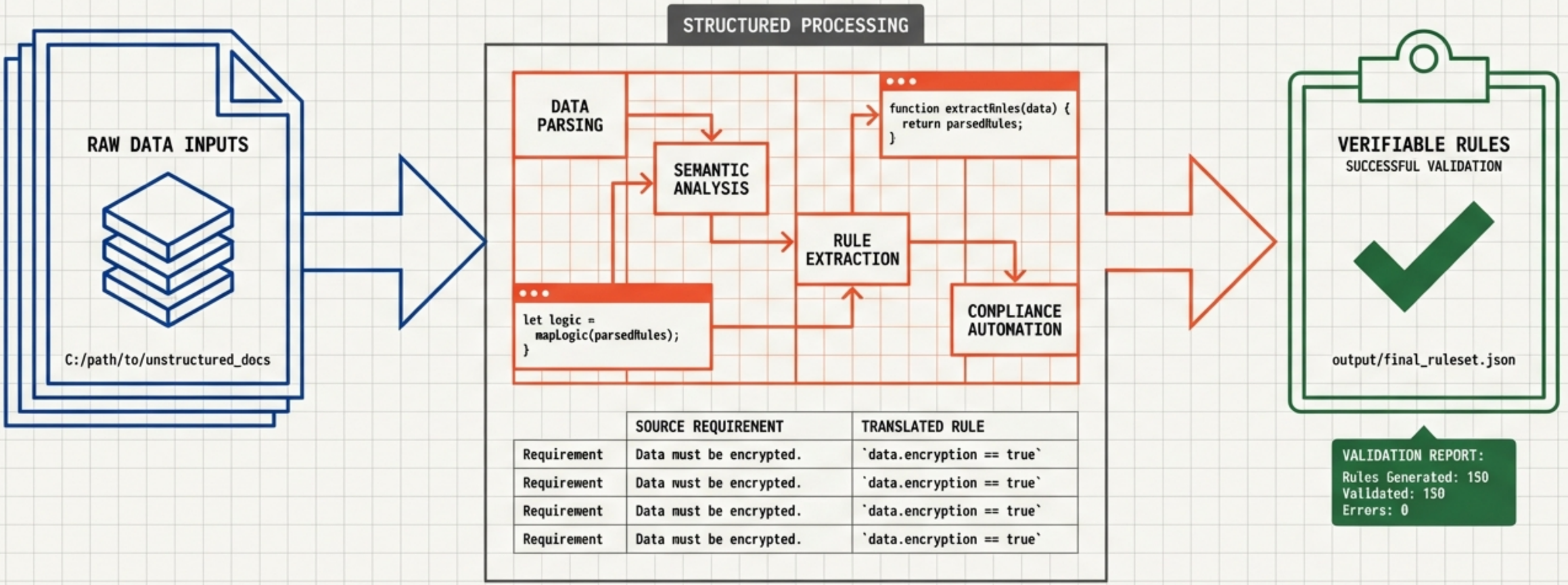
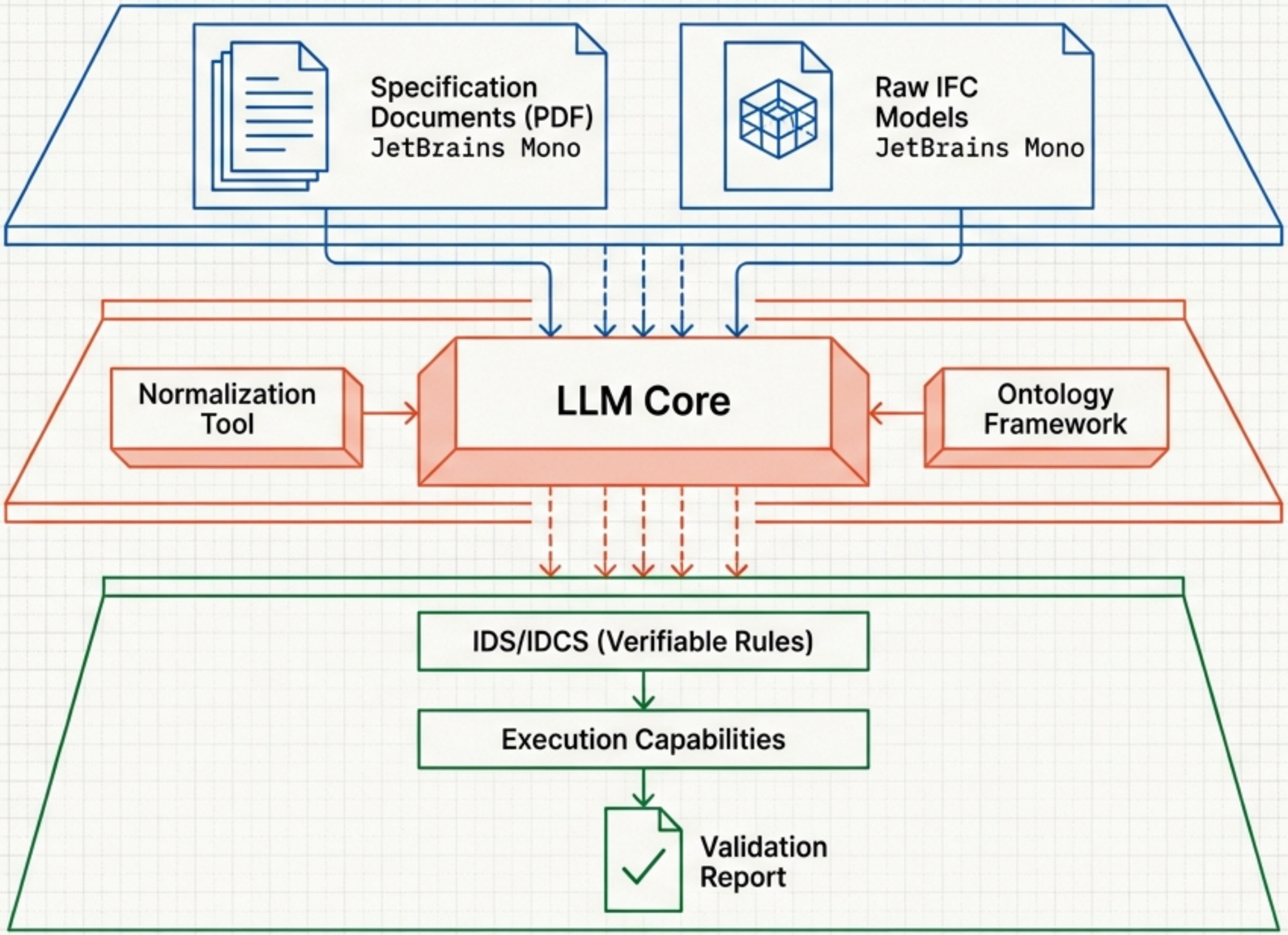
Project FFEELLAA

Automated compliance from unstructured requirements to verifiable rules.

Technical Onboarding and System Architecture Reference

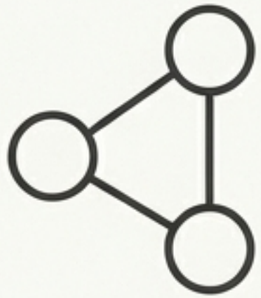


Unstructured text becomes executable logic



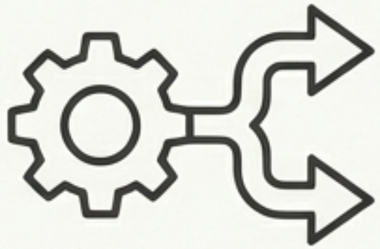
The project architecture is divided into logic, execution, and artifacts

ontologies/



Holds the core and rule ontologies formatted in Turtle.

Capabilities



Houses scripts for ICDD/IFC conversion and TTL to IDS extraction.

ids/

Stores the resulting IDS/IDCS artifacts used for active validations.

Capability breakdown: Normalization vs. Extraction

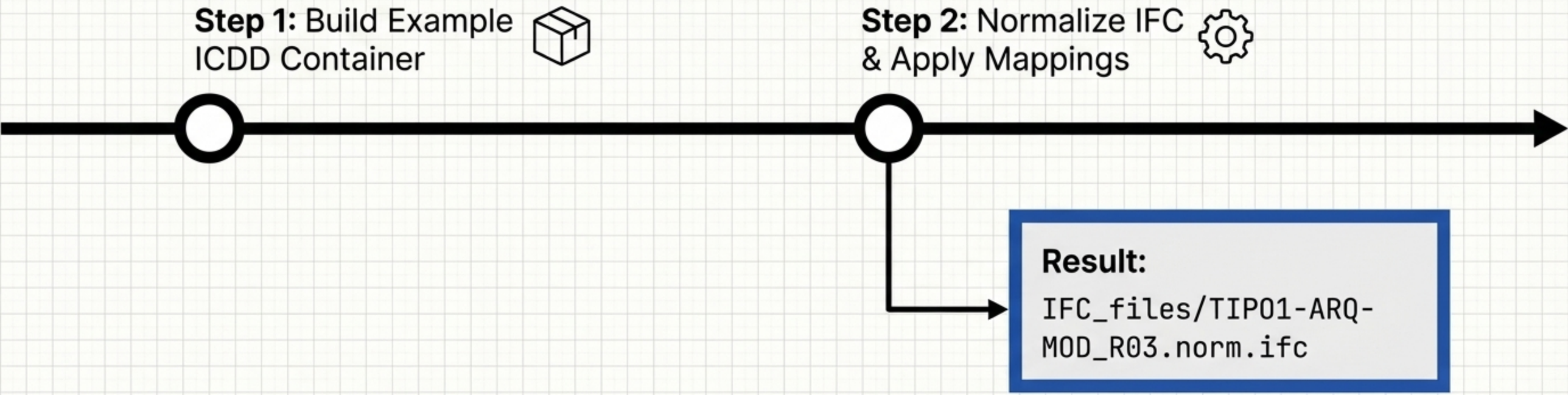
	convert_icdd_ontology.py	ttl_to_ids.py
Primary Function	Normalizes IFC & applies ICDD mappings	Filters IDS-expressible rules from Turtle.
Library Stack Stack	ICDD processing tools	RDFLib parser
Core Output	Normalized IFC	XML-based specifications

Directory: **capability/package/capability/plugin/capability/**

Dynamic matching prevents rigid namespace coupling

The TTL parser (using RDFLib) avoids hardcoding namespaces. Instead, it dynamically searches for predicates matching the localname. This ensures flexible rule extraction.

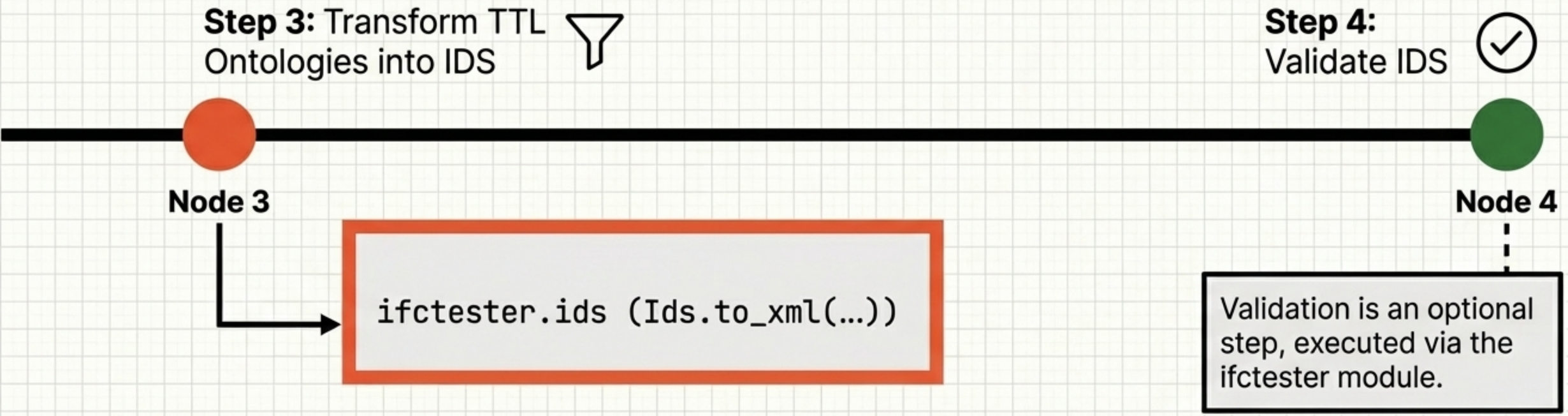
Execution Phase 1: Containerize and Normalize



Assumes virtual environment is fully configured prior to execution.

Execution Phase 2: Transformation and Validation

Validation is an optional step, executed via the ifctester module.



Verifying the TTL rule extractor with pytest

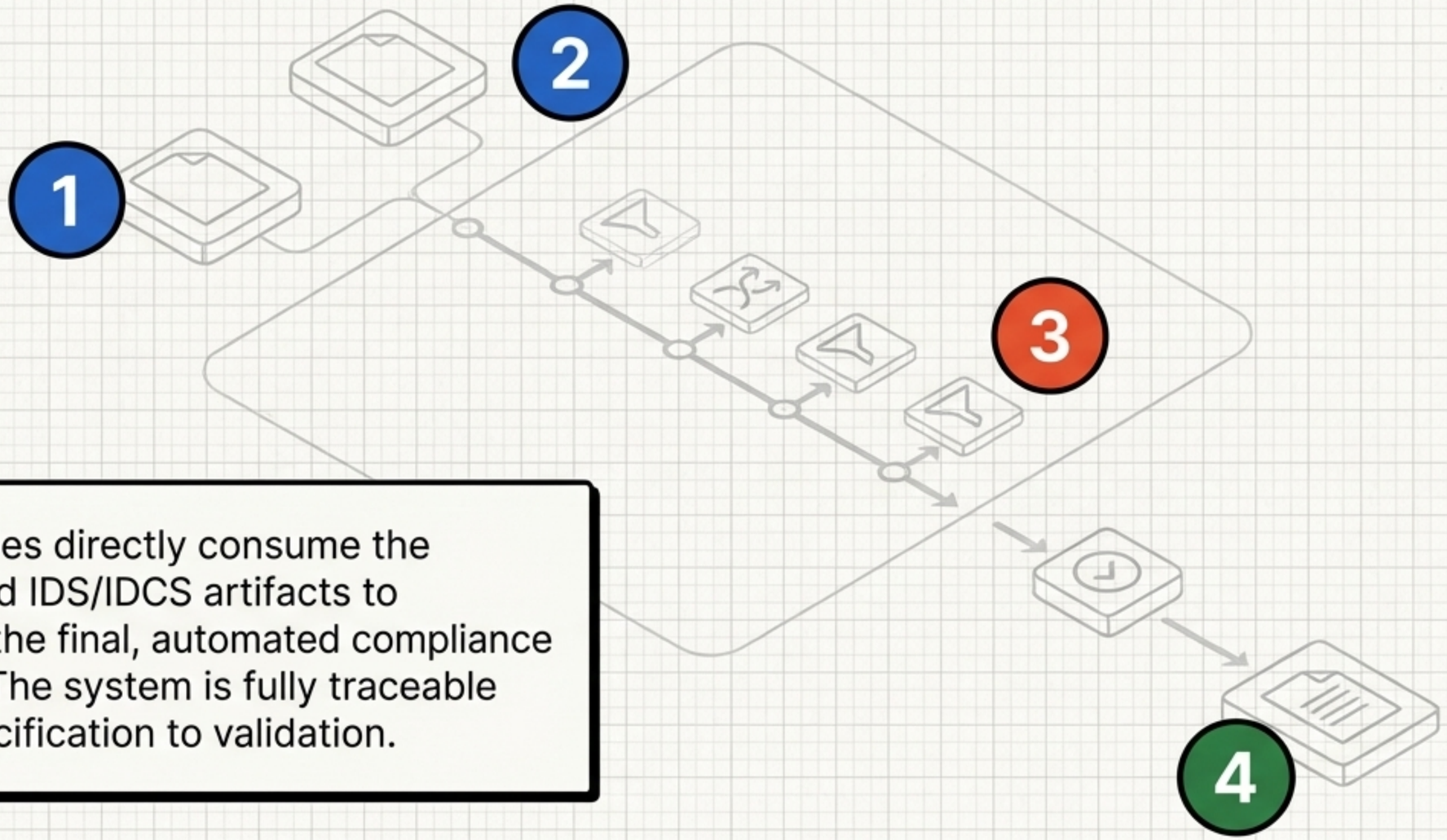
Dedicated unit tests ensure the TTL to IDS transformation accurately filters expressible rules before entering the main validation pipeline.

`tests/test_ttl_to_ids_read_fela_computable_rules.py`

`test_extract_rules` **PASSED**

This dedicated unit test verifies that the TTL to IDS transformation process correctly identifies and extracts only the expressible rules from the input ontologies, ensuring data integrity before the main validation pipeline begins.

The refinery in motion: from raw PDF to verified report



Capabilities directly consume the generated IDS/IDCS artifacts to produce the final, automated compliance reports. The system is fully traceable from specification to validation.